

Derek Wang

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TECHNICAL SKILLS

Languages: C++, Python, MATLAB, Java, JavaScript, TypeScript, SQL

Frameworks/Libraries: ROS 2, PyTorch, OpenCV, Eigen, MuJoCo, ONNX, React, Next.js, Node.js, FastAPI

Infrastructure: Docker, GitHub Actions, CMake, Git, SQLite, MongoDB, Redis, Foxglove

EXPERIENCE

Software Engineer Intern

May 2026 – August 2026

Prandtl Dynamics Inc.

Toronto, ON

- Tracked 5+ moving targets at 170 m with 20 ms latency by chaining SRP-PHAT azimuth/elevation estimation, delay-and-sum beamforming per source, and Unscented Kalman Filter + Global Nearest Neighbor tracking
- Raised acoustic drone detection to 98% mAP with 40% fewer false alarms and half the model size via EfficientAT
- Closed cross-tenant and auth gaps across 6 dashboard services with middleware that injects org-ID predicates into every query, scopes WebSocket rooms per org, and fails closed on JWT checks, enforced by CI audit gates
- Automated cleaning of 50+ hours of training audio via a Python tool localizing and beamforming array recordings
- Slashed recording setup from 10 to 2 minutes by swapping a tethered rig for a Raspberry Pi audio service over SSH

Software Engineer Intern

March 2026 – June 2026

Kenosis Technologies Inc.

Mississauga, ON

- Lowered API latency by 35% by caching posting lookups and auth sessions in Redis across Node.js microservices
- Designed MongoDB schemas and ingestion pipelines for job postings, improving refresh from daily to hourly
- Flagged suspicious applicants 60% faster by queuing LLM document review for fraud, scoring, and LMIA results

Autonomy Software Developer

September 2025 – Present

WATonomous – Student Design Team

Waterloo, ON

- Built a ROS 2 navigation stack running A* path planning over a 100x100 m costmap and Pure Pursuit control under 10 ms per cycle, replanning at 10 Hz for real-time autonomous rover navigation
- Led YOLO11s model retraining, improving detection mean average precision from 30% to 80% across 3,448 labeled images via hyperparameter tuning and augmentation redesign
- Accelerated YOLO11s inference to 30 FPS on a Jetson Nano via ONNX export and onnxslim graph simplification

PROJECTS

Mr. Clean - LLM-Orchestrated Mobile Manipulation | *Python, ROS 2, MuJoCo, PyTorch, SB3, Claude API*

- Drove 9/9 routes within 10 cm / 5° with 0 falls under a classical nav controller (A*, clamped cubic splines, footprint-inflated occupancy grid, dock navigator that replans live) with SLAM Toolbox localizing to 1.4 mm / 0.2°
- Ran a 7 min 48 s tour from one prompt and tidied 4/4 cubes in 13 tool calls under a two-level Claude agent controller (commander, nested skills agent with 6 typed skills) that refuses unknown objects at the tool boundary
- Trained an Action Chunking Transformer controller (ResNet-18 over 3 cameras, 4+4-layer encoder-decoder, 22M params) on 80 scripted demos to predict 32-step joint chunks at 20 Hz, reaching the ball in 4/10 unseen layouts
- Controlled the base with a 512-neuron, 8,688-edge FlyWire connectome subgraph as an SB3 feature extractor (4 rounds of transmitter-signed message passing), hitting 20/20 held-out point goals after behavior cloning

7-DOF Robotic Arm Motion Control | *C++17, Eigen, MuJoCo, CMake, React, TypeScript*

- Derived Lie-group kinematics and recursive Newton-Euler dynamics in C++17/Eigen, matching MuJoCo to 1e-13
- Held sub-0.2 mm tip error with zero actuator saturation over 50+ scenarios via jerk-limited 1 kHz computed torque
- Cut worst-case IK replan latency from 40.7 to 10.5 ms across 324 targets via damped least squares with 26 seeds
- Resolved redundancy via annealed null-space projection of 4 objectives and swept joint-space collision checks

Autonomous Vehicle State Estimation & Tracking | *C++, Eigen, ROS 2, Python, Docker, KITTI*

- Implemented a 15-state error-state Kalman filter fusing 100 Hz IMU with 10 Hz GPS in C++/Eigen on ROS 2
- Localized the ego vehicle to 0.558 m mean error over 8 KITTI sequences (6.6 km), cross-checked via evo
- Reached 0.886 MOTA on ground-truth detections with a CV/CA/CT IMM of UKFs and Hungarian association
- Coupled tracking to localization by world-framing detections with the filter's pose (error correlation 0.976)

EDUCATION

University of Waterloo

Expected April 2030

Bachelor of Applied Science in Computer Engineering (Co-op)